
Brain Tumour MRI Classification — Research Pipeline
Comparing Polar (PoPE) vs Rotary (RoPE) positional embeddings in Vision Transformers on 4-class MRI classification
Brain MRI Dataset
3,264 images · 4 classes
Kaggle Brain Tumour Classification
No Tumour
~900 imgs
Meningioma
~704 imgs
Glioma
~826 imgs
Pituitary
~827 imgs
Stratified Split
Train
76% · 2,476 imgs
Val
12% · 394 imgs
Test
12% · 394 imgs
Preprocessing / Augmentation
Train (augmented)
Resize 224 × 224
RandomHorizontalFlip p=0.5
RandomAffine ±15°
Normalize μ/σ
Val / Test (deterministic)
Resize 224 × 224
Normalize μ/σ
Model Zoo (4 architectures)
PoPEViT (ours)
Polar positional encoding on Q, K
ViT-base · d=512 · L=6 · h=8
RoPEViT (baseline)
1-D rotary positional embedding
ViT-base · d=512 · L=6 · h=8
DeiT-Small + PoPE/RoPE
Pretrained ViT, attn heads swapped
ImageNet weights · 22M params
ResNet-18 (CNN baseline)
Standard CNN, trained from scratch
11M params · batch-norm
Optimisation
AdamW
lr = 3×10^{-4}
weight decay = 1×10^{-2}
Cosine Annealing LR
5-epoch linear warm-up
max epochs = 30
Class-Weighted CE Loss
$w_i \propto 1 / freq(class_i)$
corrects class imbalance
Gradient Clipping
max_norm = 1.0
prevents exploding gradients
Early Stopping
patience = 8 epochs
monitor val AUROC
Evaluation
Mean AUROC (macro)
primary metric
one-vs-rest per class
Per-class AUROC
no_tumor · meningioma
glioma · pituitary
Accuracy
secondary metric
4-class top-1
Confusion Matrix
normalised 4×4
heatmap visualisation
Best Results
ResNet-18: AUROC 0.999 · PoPEViT: AUROC 0.964
① Dataset
② Split
③ Preprocessing
④ Models
⑤ Optimisation
⑥ Evaluation